

A. ENCIRCLE THE CORRECT ANSWER:

1. To "bake a cake" is an example of:
   a. problem **b.** strategy
   c. algorithm **d.** solution
2. To find a feasible solution to a problem, the first step is to:
   a. establish starting point **b.** find available solutions
   c. create a strategy **d.** identify and analyze the problem
3. Step by step solution of a problem in simple language is called:
   a. Problem Solving **b.** Algorithm
   c. Flowchart **d.** Data Structure
4. _____ shows the logic of program graphically.
   a. Data Structure **b.** Graph
   c. Algorithm **d.** Flowchart
5. _____ Symbol is used for input/output in flowchart.
   a. Triangle **b.** Square
   c. Parallelogram **d.** Oval
6. Elements of _____ data structure are not connected sequentially.
   a. Array **b.** Graph
   c. Queue **d.** Stack
7. _____ stores data in hierarchal manner.
   a. Stack **b.** Queue
   c. Array **d.** Tree
8. When that data is Pushed in stack, it means that data is:
   a. inserted **b.** deleted
   c. sorted **d.** edited
9. In binary tree, each child can have maximum:
   a. one node **b.** two nodes
   c. three nodes **d.** four nodes
10. Traversing an array means accessing:
    a. first element **b.** last element
    c. any specific element **d.** each and every element of the array

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A. ENCIRCLE THE CORRECT ANSWER:

1. A computer program is a collection of:
   a. Tasks b. Instructions
   c. Computers d. Programmers
2. High-level languages have syntax that is:
   a. Easily readable by humans b. Easily readable by machines
   c. Easily readable by both d. None of the above
3. Low-level languages have syntax that is:
   a. Easily readable by humans b. Easily readable by machines
   c. Easily readable by both d. None of the above
4. The primary characteristic of a compiler is to:
   a. Translate codes line-by-line
   b. Translate low-level code to machine language
   c. Detect logical errors
   d. Translate codes all at once
5. The primary characteristic of an interpreter is to:
   a. Translate codes line-by-line
   b. Translate low-level code to machine language
   c. Detect logical errors
   d. Translate codes all at once
6. An Integrated Development Environment facilitates a programmer to:
   a. Edit source code b. Complete and highlight syntaxes
   c. Debug and compile codes d. All of the above
7. All errors, detected by users are typically:
   a. Syntax Errors b. Semantic Errors
   c. Run- Time Errors d. Logical Errors
8. Allowed names for declaring a variable:
   a. Can contain whitespaces
   b. Can be one of the reserved words
   c. Can contain letters, digits and underscores
   d. Can be the same as its data type

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9. A bool data can store following type of value:
   a. Numbers b. Strings
   c. Fractional numbers c. True or false
10. Which data type occupies the most space in memory?
   a. Character b. Integer
   c. Floating point d. Double floating point

B. RESPOND THE FOLLOWING:

1. What is computer program?
2. List five common high-level languages used and describe their purpose.
3. Using the rules of naming variable, develop ten meaningful and valid variable names.
4. Write and two differences between machine and assembly language..
5. What are Strings in C++?
6. What is the difference between declaring and initializing a variable?
7. What is the difference between source code and object code?
8. List any four advantages of using an IDE.



1. In groups, students should learn to download, install and configure Dev C++.
2. Teacher demonstrates the use of IDE and its features as given in this unit. Also explains the use of variable and constants.

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A. ENCIRCLE THE CORRECT ANSWER:

i) The C++ header file _____ contains function prototype for the standard Input and Output functions.

a. <iomain.h> b. <iostream>
c. <fstream.h> d. <cstdio.h>

ii) Which operator is used for input stream?

a. > b. <<
c. >> d. <

iii) gets stands for _____.

a. get stream b. get string
c. get str d. get std.

iv) getch() and getche() are included in _____ header file.

a. <cstdio.h> b. <conio.h>
c. <stdlib.h> d. <stdio.h>

v) Which operator is used for logical AND operation?

a. & b. &&
c. || d. !

vi) Which of the following operator is correct to compare two values of variables?

a. = b. <=
c. == d. both b and c

vii) Which of the following needs pressing Enter Key from the keyboard?

a. getch( ) b. getche( )
c. getchar( ) d. gets( )

viii) != operator belong to which type of operator.

a. Relational b. Logical
c. Arithmetic d. None of these

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ix) Which operator add the first operand to the second operand and gives the result to first operand.

a. *= b. +=

c. ++ d. +

x) `cout << 12-6/2;` What will be the result on screen?

a. 3 c. 6

c. 9 d. 12

B. RESPOND THE FOLLOWING:

1. Use \a and \r both escape sequences in a program.
2. How many types of **comment** statements are used in C++?
3. Differentiate between Arithmetic operators and Relational operators.
4. Write a program in C++ and use all arithmetic assignment operators.
5. What is basic difference between Assignment operator and Equal to operator.
6. What is the basic difference between \n and \t ?
7. Get the output of following program.

```
#include <iostream>
using namespace std;
int main (  )
{
    int a = 27;
    cout << "a is " << a << endl;
    cout << "a is now" << a++ << endl;
    cout << "a is now " << a << endl;
    cout << "a is now " << --a << endl;
    cout << "a is now " << a << endl;
return 0;
}
```

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- A **"continue"** statement causes the loop to skip the remaining statements of its body and immediately transfers control to the top of the loop.
- A **"goto"** statement jumps or transfers control unconditionally from the **"goto"** to a labeled statement in the same function.
- A **"return"** statement terminates the execution of a function and transfers program control to the statement just after the function call statement in the calling function.
- The **exit()** is used to terminate a C++ program.



1. Encircle the correct answer:

i) Loop within a loop is called _____ loop.
   a. inner b. outer
   c. enclosed d. nested

ii) "case" and _____ are also part of "switch" statement.
   a. have b. default
   c. for d. if

iii) "for" Loop expression has _____ parts.
   a. one b. two
   c. three d. four

iv) exit( ) function is used to _____.
   a. close function b. close loop
   c. close program d. close switch

v) "continue" statement takes control to the _____ .
   a. top of loop b. end of loop
   c. top of function d. end of function

vi) In "goto" statement label is followed by _____ character.
   a. colon (:) b. semi colon (;)
   c. single quote (') d. double quote (")

vii) To send value to the calling function we use _____ statement.
   a. throw b. return
   c. send d. back

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viii) "break" statement is used with _____.

a. if b. switch
c. for d. while

ix) Using "else" is _____ with "if" statement.

a. prohibited b. advised
c. compulsory d. optional

x) "if" and loop expressions use _____ operators to test condition.

a. arithmetic b. relational
c. insertion d. bitwise

2. Answer the following questions:

1. What is the purpose of "**default**" statement in C++?
2. Can we use "**while**" loop in place of "**for**" loop, if yes then how?
3. What is the main difference between **while** and **do while** loops?
4. Write the function of **for** loop.
5. Why we make block of statements using braces?
6. Which data type variables can be used in "**switch**" statement?
7. What is the purpose of jump statements?
8. Write the purpose of following statements:
   a. else if
   b. switch
   c. goto
   d. exit

3. Match the column:

S.no	A	S.no	B
(a)	if	(i)	Relational operators
(b)	loop	(ii)	break
(c)	Conditional expression	(iii)	switch
(d)	Loops and switch	(iv)	operator
(e)	do	(v)	iteration
(f)	>>	(vi)	else
(g)	case	(vii)	while

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A. ENCIRCLE THE CORRECT ANSWER:

1. The functions that are defined by the programmer are called:
   a. Built-In function b. User-defined-function
   c. Subfunction d. Function
2. A programmer creates a function for a particular task and the programmer wants to include that function in program. Which extension is required to save that function?
   a. .obj b. .h
   c. .cpp d. .exe
3. In C++, int main( ) returns which data type value by default?
   a. float b. int
   c. char d. double
4. The parameters specified is the function header are called:
   a. formal parameters b. actual parameters
   c. default parameters d. command line parameters
5. The word "prototype" means:
   a. Declaration b. Calling
   c. Definition d. Both a & b
6. The function prototype consists of:
   a. Name of function
   b. the parameters are passed to the function
   c. The value return from function
   d. All of these
7. All variables declared in function definition are called
   a. Local variable b. Instance variable
   c. Global variable d. Static variable
8. Which are not the built-In function?
   a. sqrt() b. time()
   c. exp() d. sin()

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A. ENCIRCLE THE CORRECT ANSWER:

i) The feature of Scratch is _____.
   a. It is a visual
   b. Its free forever
   c. No need to remember codes
   d. All of above

ii) In Scratch, the character which moves on the Stage is called a _____.
   a. Sprite
   b. Command
   c. Script
   d. Event

iii) Repeat 10, forever and if … then codes are available in _____.
   a. Motion
   b. Control
   c. Look
   d. Sensing

iv) The Looks of Sprite can be changed by using _____.
   a. Backdrop Tab
   b. Costume Tab
   c. Script Tab
   d. Control Tab

v) A Scratch Program is at least made up with _____.
   a. Many Sprites
   b. One Sprite and Code
   c. One Sprite
   d. One Sprite and Backdrop

vi) To change the position of Sprite on screen, we use _____.
   a. Coordinates
   b. Stage Information
   c. Command Pallets
   d. Costume Tab

vii) 

can be found under _____.
   a. Look
   b. Motion
   c. Sound
   d. Control

viii) Turn command, turns Sprite to specified:
   Coordinate
   c) Steps
   Degree
   d) Seconds

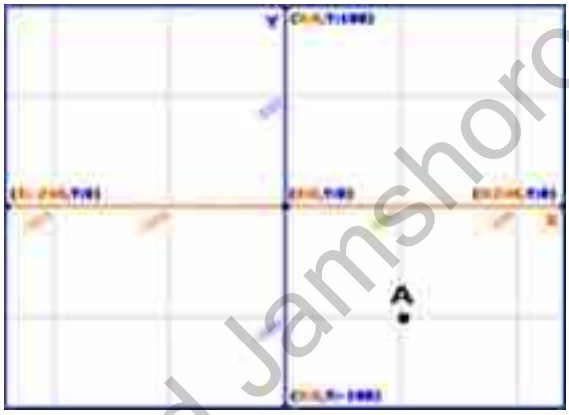
ix) This 

command is available in _____.
   a. Event
   c. Control
   c. Motion
   d. Looks

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x) In the given picture identify the x and y coordinates (positions) of Point – A.

a. x=100, y=-100
b. x=-100, y=100
c. x=100, y=100
d. x=-100, y=-100



B. RESPOND THE FOLLOWING:

1. Explain the following:
   i. Script
   ii. Sprite
   iii. Backdrop
2. State the difference between repeat 10 and forever commands.
3. Write the use of the following codes: forever, wait, say, play sound, go to x, y
4. What is the difference between using Scratch online and offline?
5. In your notebook mark the color of each pallet in Script Tab. One is done for you. How are these colors helpful for the user?

	Pen

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